

2026 复旦插班生数学

终章

例题 0.1 $f(x,y,z)$ 是三元函数，对于任意的 $\alpha_1,\alpha_2,\alpha_3$，恒有 $f(\alpha_1 x_1,\alpha_2 x_2,\alpha_3 x_3)=\phi(\alpha_1,\alpha_2,\alpha_3)f(x_1,x_2,x_3)$, 求证 $\exists d_1,d_2,d_3$ 使得 $\phi(\alpha_1,\alpha_2,\alpha_3)=\alpha_1^{d_1}\alpha_2^{d_2}\alpha_3^{d_3}$.

例题 0.2 $f(0)=0, f'(0)=f'(1)=0$, 求证

$$\left|\int_0^1 f(x)\,dx\right| \le \frac{1}{3\sqrt{5}}\left(\int_0^1 (f''(x))^2\,dx\right)^{\frac{1}{2}}$$

证明

$$\begin{aligned}\int_0^1 f(x)\,dx &= -\int_0^1 f(x)d(1-x) = \int_0^1 f'(x)(1-x)\,dx - (1-x)f(x)\Big|_0^1\\ &= \int_0^1 f'(x)(1-x)\,dx = \int_0^1 f'(x)d\left(x-\frac{x^2}{2}+C\right)\\ &= \left(x-\frac{x^2}{2}+C\right)f'(x)\Big|_0^1 - \int_0^1\left(x-\frac{x^2}{2}+C\right)f''(x)\,dx\\ &= \int_0^1\left(\frac{x^2}{2}-x+C\right)f''(x)\,dx\end{aligned}$$

故

$$\left|\int_0^1 f(x)\,dx\right| \le \sqrt{\int_0^1\left(\frac{x^2}{2}-x+C\right)^2 dx}\sqrt{\int_0^1 [f''(x)]^2\,dx}$$

下面确定系数 C

$$\begin{aligned}\int_0^1\left(\frac{x^2}{2}-x+C\right)^2 dx &= \int_0^1\left(\frac{x^2}{2}-x\right)^2 + 2C\left(\frac{x^2}{2}-x\right)+C^2\,dx\\ &= C^2-\frac{2}{3}C+\frac{2}{15} \ge \frac{4ac-b^2}{4a} = \frac{\frac{8}{15}-\frac{9}{4}}{4} = \frac{1}{45}\end{aligned}$$

故取 $C=-\frac{b}{2a}=\frac{1}{3}$ 时最佳系数为 $\sqrt{\frac{1}{45}}=\frac{1}{3\sqrt{5}}$.